

Analog Integrated Circuit Design

Cadence Virtuoso Tools

Lab 02

Common Source Amplifier

Technology	Gain	Bias Current
0.18 μm CMOS	$ A_v = 10$	$I_{DQ} = 80 \mu\text{A}$

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1 1

Part

1: Sizing Chart

1.1 Design Specifications

The target specifications for the 0.18 μm CMOS technology are:

Table 1. Design specifications.

Parameter	Value
DC Gain $ A_v $	10
Supply V_{DD}	1.8 V
Current consumption I_{DQ}	80 μA
Channel length L	2 μm

This version of the generator creates a random integer. It can deal with very large integers up to a few thousand digits.

Result

80

Lower Limit 11
Upper Limit 100

Generate Clear

(a) Random number generator showing bias current = 80 μA .

This version of the generator creates a random integer. It can deal with very large integers up to a few thousand digits.

Result

10

Lower Limit 5
Upper Limit 10

Generate Clear

(b) Random number generator showing gain = 10.

Figure 1. Randomly assigned design parameters for this report.

1.2 Hand Analysis: Design Equations

1.2.1 Load Resistor R_D

To maximise output signal swing, the common-mode (CM) output is placed at $V_{DD}/2$, which means the DC voltage drop across R_D equals $V_{DD}/2$:

$$V_{R_D} = \frac{V_{DD}}{2} = \frac{1.8}{2} = 0.9 \text{ V} \quad (1)$$

Given the DC bias current $I_{DQ} = 80 \mu\text{A}$:

$$R_D = \frac{V_{R_D}}{I_{DQ}} = \frac{0.9}{80 \times 10^{-6}} = \boxed{11.25 \text{ k}\Omega} \quad (2)$$

1.2.2 Required V_Q^*

The small-signal voltage gain of the resistively loaded CS amplifier (ignoring r_o for now) is

$$|A_v| \approx g_m R_D = \frac{2I_D}{V^*} R_D = \frac{2V_{R_D}}{V^*} \quad (3)$$

where $V^* = 2I_D/g_m$ is the effective overdrive (equal to V_{ov} only for a square-law device). Rearranging:

$$V_Q^* = \frac{2V_{R_D}}{|A_v|} = \frac{2 \times 0.9}{10} = \boxed{0.18 \text{ V}} \quad (4)$$

1.2.3 Transistor Sizing: Width W

Because $I_D \propto W$ at any fixed V_{GS} (irrespective of the operating model), the sizing chart generated with $W_{\text{ref}} = 10 \mu\text{m}$ gives a reference current I_{DX} at $V^* = V_Q^*$. The required width is then found by ratio and proportion:

$$\frac{W}{10 \mu\text{m}} = \frac{I_{DQ}}{I_{DX}} \implies W = 10 \mu\text{m} \times \frac{I_{DQ}}{I_{DX}} \quad (5)$$

1.3 Simulation Results: Sizing Chart

1.3.1 V^* and V_{ov} vs. V_{GS} ($V_{th} = 0.383 \text{ V}$)

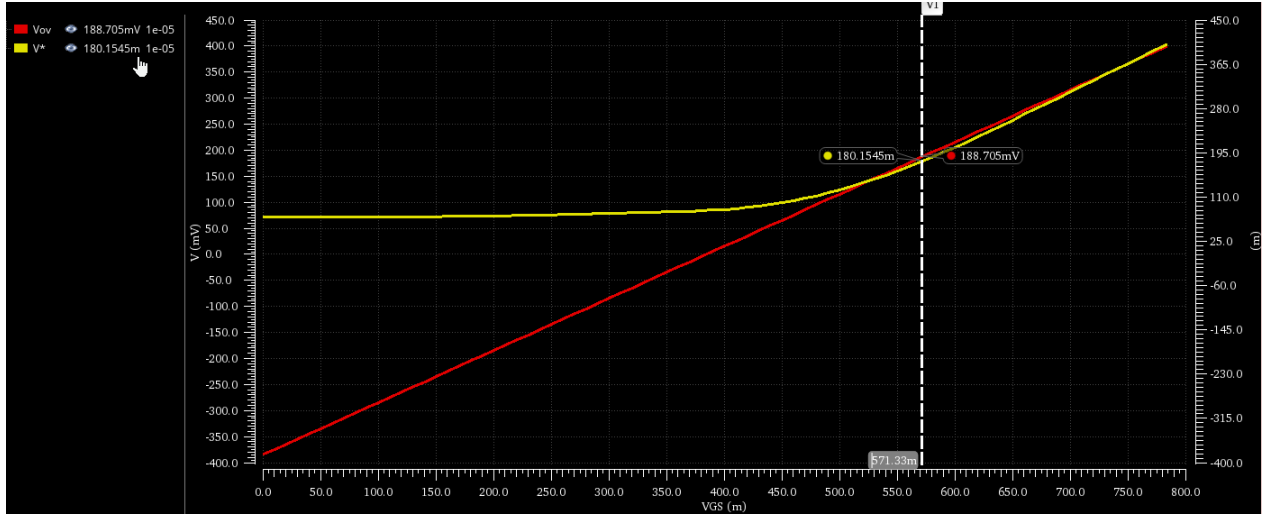


Figure 2. $V^* = 2I_D/g_m$ (Yellow) and $V_{ov} = V_{GS} - V_{TH}$ (Red) vs. V_{GS} for the NMOS with $W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$.

1.3.2 I_D , g_m , g_{ds} vs. V_{GS}

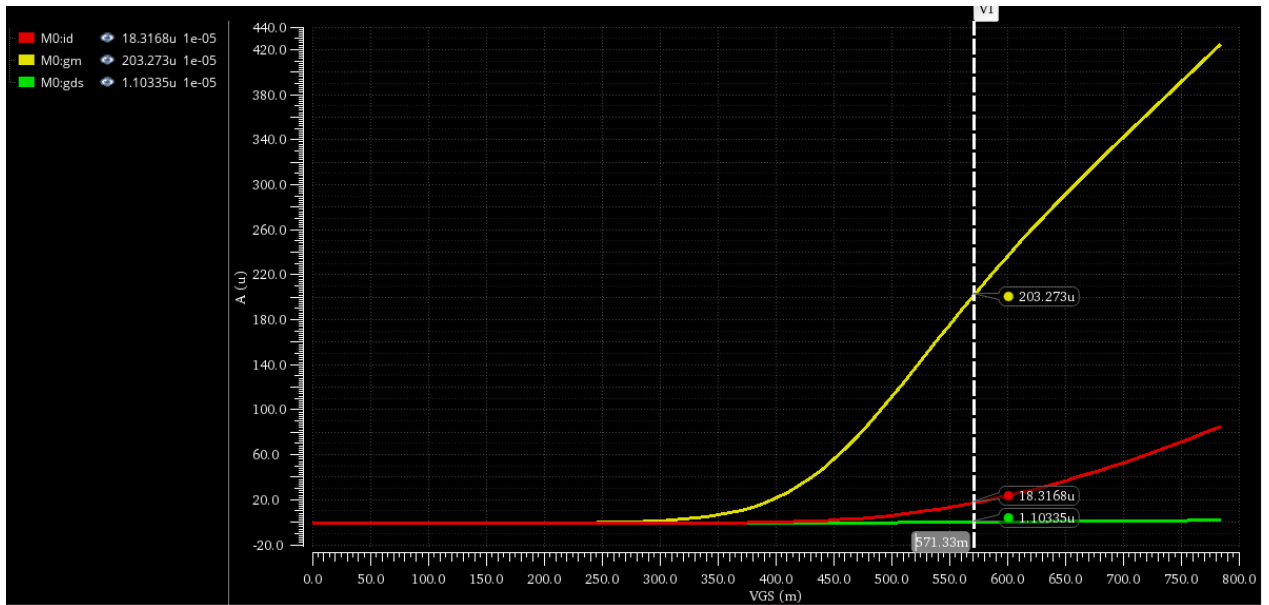


Figure 3. I_D , g_m , and g_{ds} vs. V_{GS} for the reference NMOS ($W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$, $V_{DS} = V_{DD}/2$).

From the sizing chart (Figure 2), at $V_Q^* = 0.18 \text{ V}$ we read:

Table 2. Values read from the sizing chart at $V_Q^* = 0.18\text{ V}$, $W_{\text{ref}} = 10\text{ }\mu\text{m}$, $L = 2\text{ }\mu\text{m}$.

Quantity	Symbol	Value
Reference drain current	I_{DX}	$18.32\text{ }\mu\text{A}$
Reference transconductance	g_{mX}	$203.27\text{ }\mu\text{S}$
Reference output conductance	$g_{ds,X}$	$1.103\text{ }\mu\text{S}$
Overdrive voltage	$V_{ov,Q}$	188.71 mV
Gate-source bias voltage	V_{GSQ}	571.33 mV

$$W = 10\text{ }\mu\text{m} \times \frac{80\text{ }\mu\text{A}}{I_{DX}} \approx \boxed{43.67\text{ }\mu\text{m}} \quad (6)$$

1.3.3 Scaled Small-Signal Parameters

Since $g_m \propto W$ and $g_{ds} \propto W$ at constant V_{ov} :

$$g_{mQ} = g_{mX} \times \frac{W}{10\text{ }\mu\text{m}} \approx 887.68\text{ }\mu\text{S} \quad (7)$$

$$g_{ds,Q} = g_{ds,X} \times \frac{W}{10\text{ }\mu\text{m}} \approx 4.82\text{ }\mu\text{S} \implies r_{oQ} = \frac{1}{g_{ds,Q}} \approx 207.6\text{ k}\Omega \quad (8)$$

1.3.4 Gain Verification

The actual gain including r_o :

$$A_v = -g_m(R_D \parallel r_o) = -887.68\text{ }\mu \times \frac{11.25\text{ k} \times 0.2076\text{ M}}{11.25\text{ k} + 0.2076\text{ M}} = \boxed{-9.5 \approx -10} \quad (9)$$

The computed gain $|A_v| = 9.5 \approx 10$ matches the specification. The parameters are **correct** and ready for layout.

Table 3. Final design parameters from Part 1 sizing chart.

Parameter	Symbol	Value
Transistor Width	W	$43.67\text{ }\mu\text{m}$
Channel Length	L	$2\text{ }\mu\text{m}$
Transconductance	g_m	$887.68\text{ }\mu\text{S}$
Output Conductance	g_{ds}	$4.82\text{ }\mu\text{S}$
Output Resistance	r_o	$207.6\text{ k}\Omega$
Load Resistor	R_D	$11.25\text{ k}\Omega$
Gate-Source Bias Voltage	V_{GSQ}	571.33 mV

2.1.1 DC Operating Point

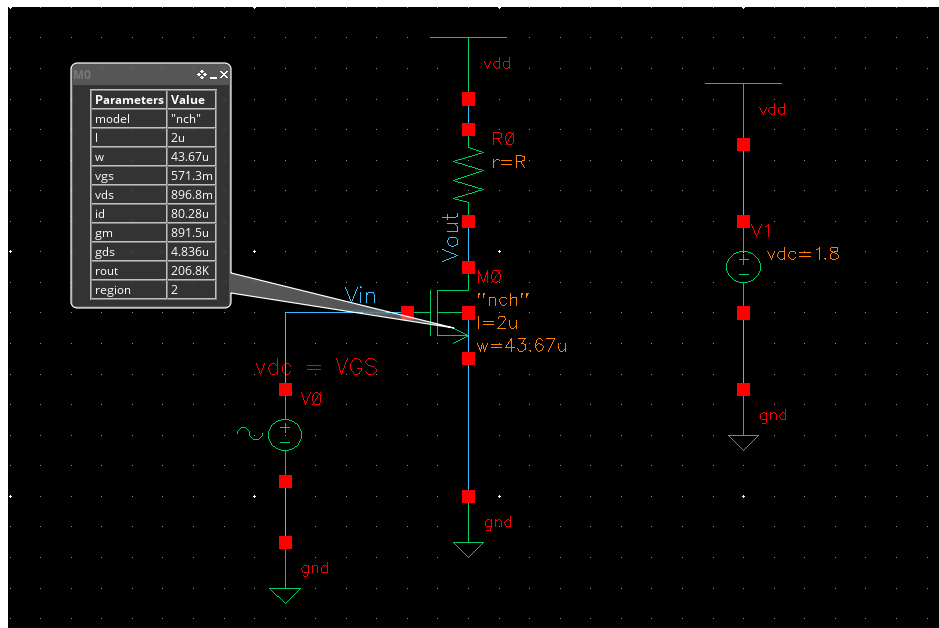


Figure 4. DC operating point annotation from Cadence Virtuoso.

Table 4. Comparison of key parameters between the sizing chart (Part 1) and DC operating point simulation (Part 2).

Parameter	Part 1 (Sizing Chart)	Part 2 (DC Simulation)	% Error
V_{GS}	571.33 mV	571.3 mV	—
I_D	80 μ A	80.28 μ A	0.35%
g_m	887.68 μ S	891.5 μ S	0.43%
g_{ds}	4.82 μ S	4.836 μ S	0.33%
r_o	207.6 k Ω	206.8 k Ω	0.39%
R_D	11.25 k Ω	11.25 k Ω	0.00%
W	43.67 μ m	43.67 μ m	0.00%
L	2 μ m	2 μ m	0.00%



Comments & Analysis

1) DC OP comparison with Part 1 (chart-based design):

Since the design used a chart extracted directly from simulation with the same technology and the same L , the operating point from the DC simulation matches the chart values closely. The drain current is approximately 80 μ A, confirming that the width scaling by ratio and proportion was correct. The region code reads 2 (saturation), as required for a linear amplifier.

2) r_o vs. R_D – is ignoring r_o justified?

From the DC OP, $r_{oQ} = 1/g_{ds,Q}$. At $L = 2 \mu$ m, r_o is typically much larger than $R_D = 11.25 \text{ k}\Omega$, so the error introduced by ignoring r_o in the gain formula $|A_v| \approx g_m R_D$ is small (a few percent). However, if minimum L were used (200 nm), r_o would drop significantly, and the parallel combination $R_D || r_o$ would be noticeably smaller than R_D alone – the assumption

would *not* be justified.

2.1.2 Intrinsic Gain and Amplifier Gain

The **intrinsic gain** of the transistor is defined as:

$$A_{\text{int}} = -g_m r_o = \frac{g_m}{g_{ds}} \approx -184 \quad (10)$$

This is the maximum possible voltage gain achievable from a single transistor (when $R_D \rightarrow \infty$, i.e., current-source load).

The **amplifier gain** with a resistive load is:

$$|A_v| = g_m (R_D \parallel r_o) = g_m \frac{R_D r_o}{R_D + r_o} \approx 9.5 \quad (11)$$

Since $r_o \gg R_D$ in the long-channel case, we have $R_D \parallel r_o \approx R_D$ and:

$$|A_v| \approx g_m R_D \ll g_m r_o = A_{\text{int}} \quad (12)$$

The amplifier gain is therefore **much less** ($\ll$) than the intrinsic gain in this design, because the resistive load R_D is much smaller than r_o .

2.1.3 AC Analysis Results

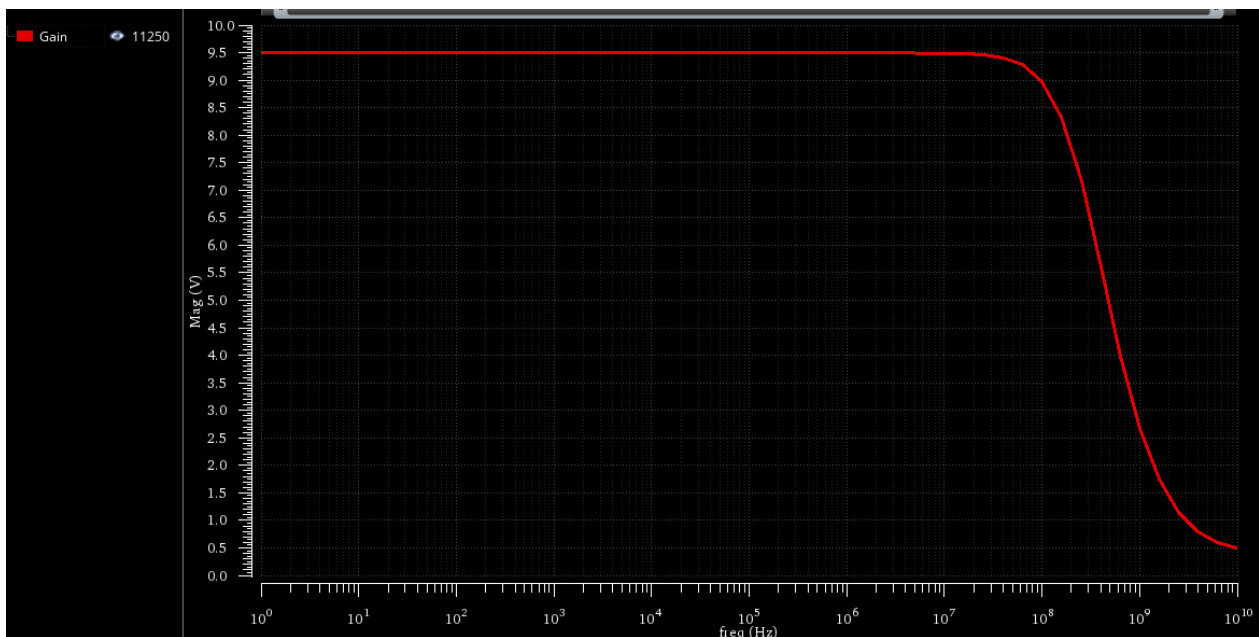


Figure 5. AC gain magnitude vs. frequency (1 Hz to 1 GHz).

ITI:LAB02_2:1	Gain		
ITI:LAB02_2:1	ymax(mag(VF("/Vout")))	9.512	

Figure 6. Annotated DC gain value from Cadence.

Comments & Analysis

The simulated DC gain from AC analysis is annotated as ≈ 9.578 (approximately 10), which agrees with the specification within reasonable tolerances of the chart-based design methodology.

The gain rolls off at high frequency due to the parasitic capacitances at the output node (C_{gd} , C_{db} , and any load capacitance).

2.2 Gain Non-Linearity

2.2.1 DC Sweep: V_{OUT} vs. V_{IN}

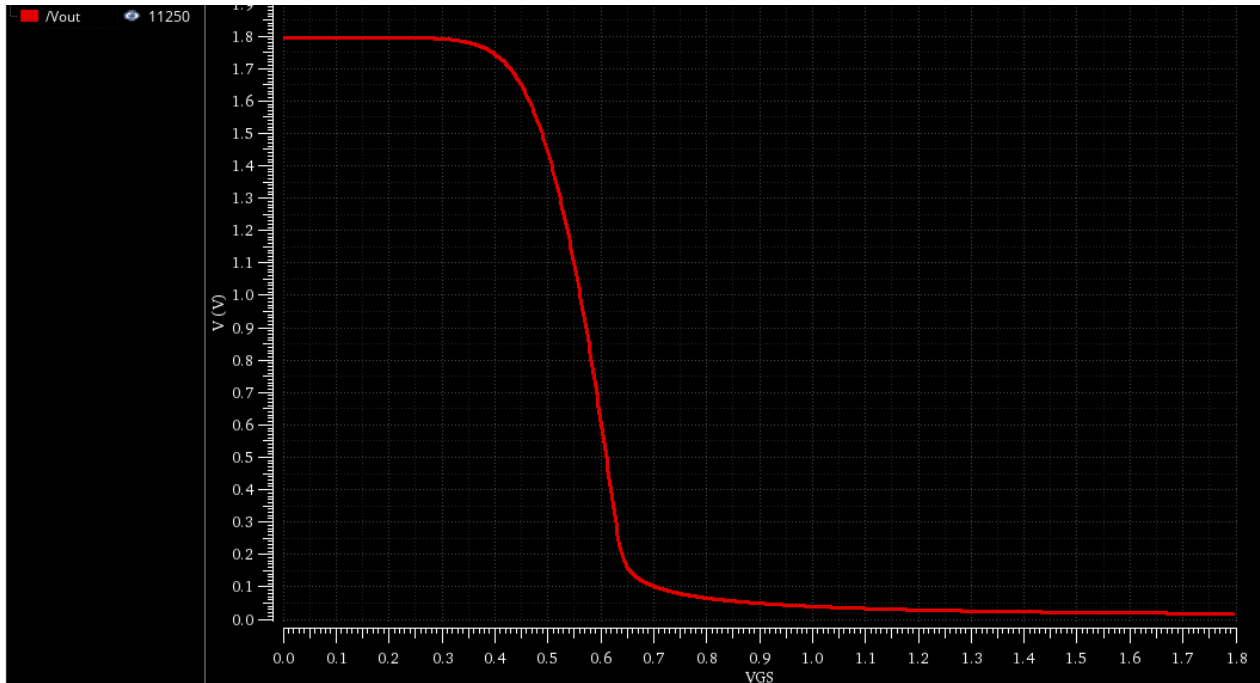


Figure 7. V_{OUT} vs. V_{IN} from DC sweep (0 to V_{DD} , 2 mV step).

Comments & Analysis

Is the V_{OUT} - V_{IN} relation linear?

No. The transfer characteristic is not linear. This is expected because:

- **Cut-off** ($V_{IN} < V_{th}$): the transistor is in cut-off; $V_{OUT} = V_{DD}$ (no current, no drop across R_D).
- **Saturation** ($V_{IN} \lesssim V_{GS} + V_{th}$): maximum slope $\approx -|A_v|$; approximately linear.
- **Triode** ($V_{IN} > V_{GS} + V_{th}$): V_{DS} drops, gain collapses, $V_{OUT} \rightarrow 0$.

The relationship is only *quasi-linear* over a small range of V_{IN} around the bias point V_{GSQ} .

2.2.2 Derivative: Small-Signal Gain vs. V_{IN}

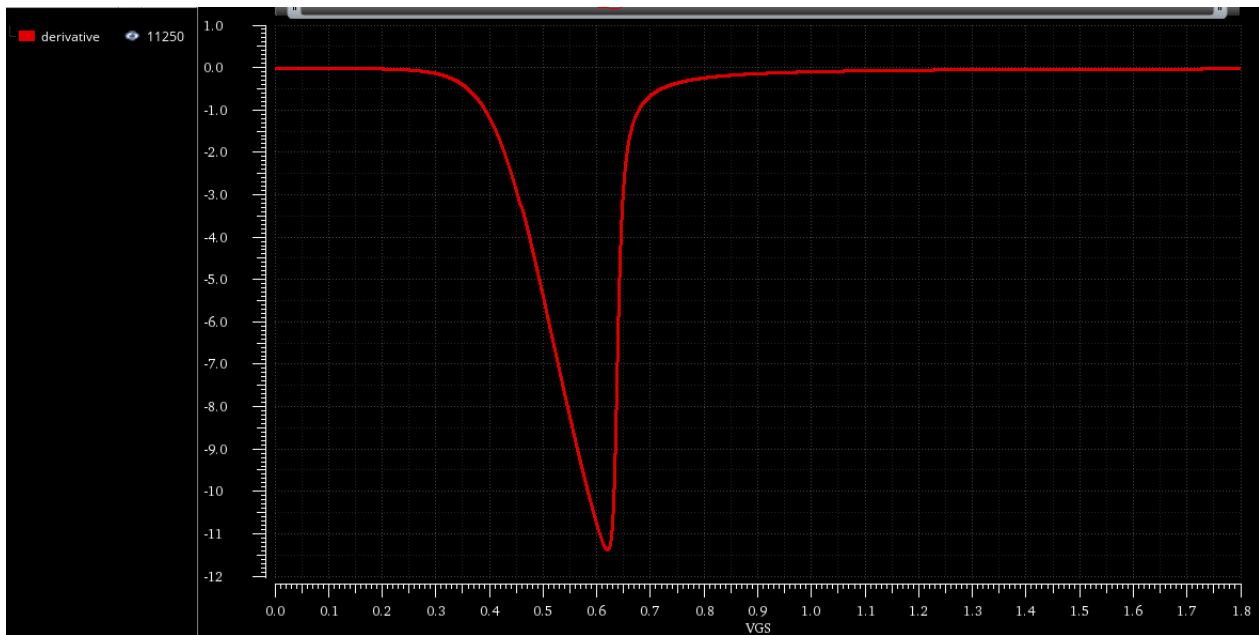


Figure 8. Derivative dV_{OUT}/dV_{IN} (instantaneous small-signal gain) vs. V_{IN} .

Comments & Analysis

Is the gain linear (independent of input)?

No. The plot of dV_{OUT}/dV_{IN} vs. V_{IN} shows that the gain varies significantly with the input voltage. This is a direct consequence of the non-linear I_D - V_{GS} relationship of the MOSFET:

- $g_m = \partial I_D / \partial V_{GS}$ is itself a function of V_{GS} (proportional to V_{ov} for square-law; saturates for short-channel).
- Since gain $= -g_m R_D$, any variation in g_m translates directly to gain variation.
- The gain is approximately constant only in a narrow range around the DC bias point.

2.2.3 Transient Simulation

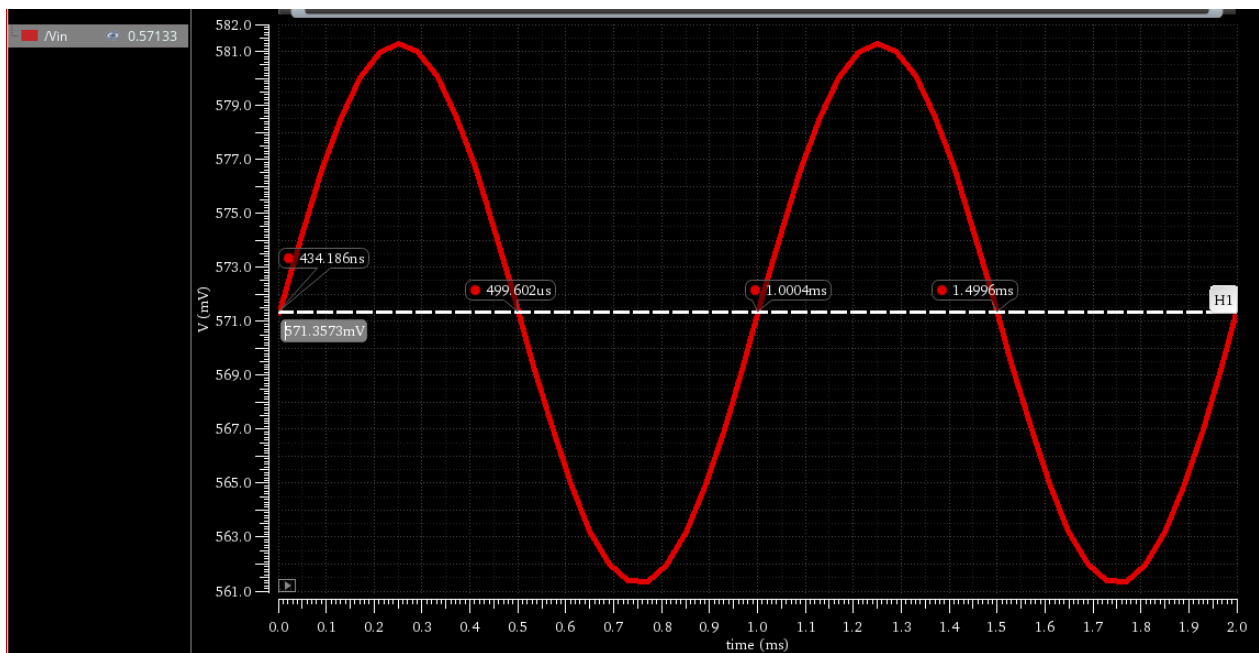


Figure 9. Input voltage V_{IN} vs. time.

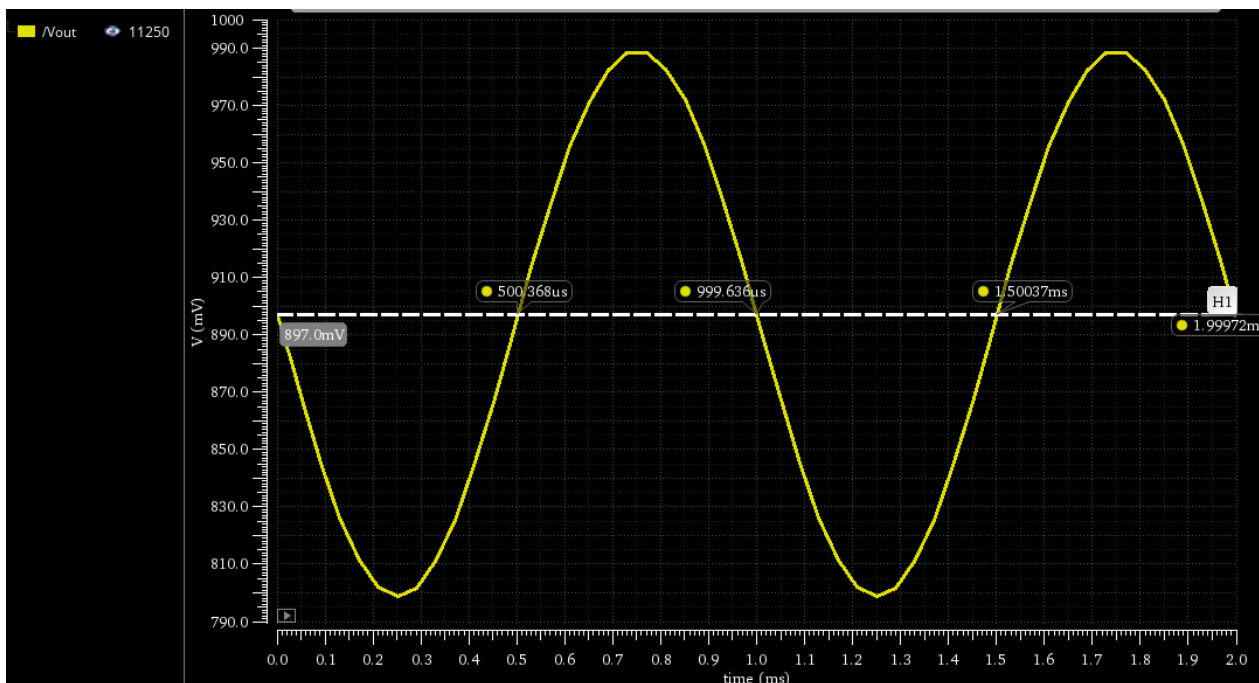


Figure 10. Output voltage V_{OUT} vs. time.

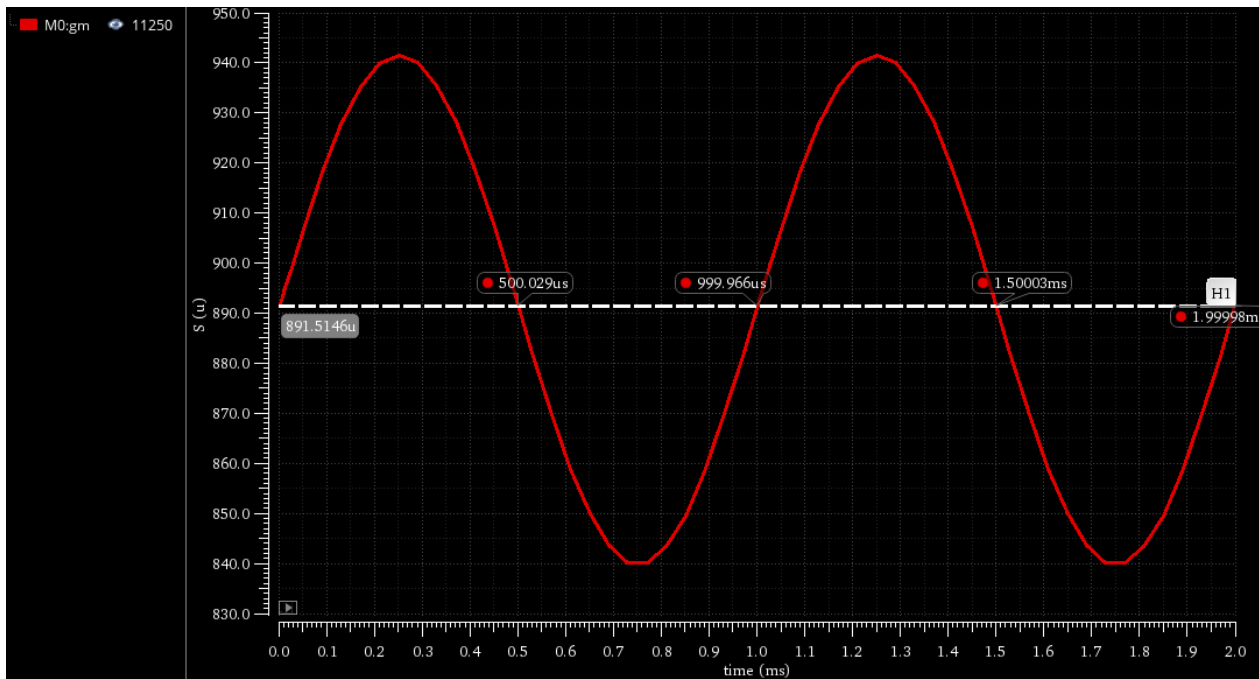


Figure 11. g_m vs. time under a 1 kHz, 10 mV sinusoidal input superimposed on V_{GSQ} .

Comments & Analysis

Does g_m vary with the input signal?

Yes. The g_m vs. time plot shows that g_m oscillates sinusoidally around its DC bias value, in phase with the input sinusoid. This occurs because g_m depends on the instantaneous V_{GS} , which is the DC bias point plus the AC signal.

What does this mean? Is this amplifier linear?

A varying g_m means the instantaneous gain $A_v(t) = -g_m(t)R_D$ is time-varying. This introduces harmonic distortion: even a pure sinusoidal input produces an output containing harmonics at $2f$, $3f$, etc. For a 10 mV amplitude the distortion is small, but it becomes severe for large signals. This amplifier is therefore **not linear** in the strict sense.

2.3 [Optional] Maximum Gain

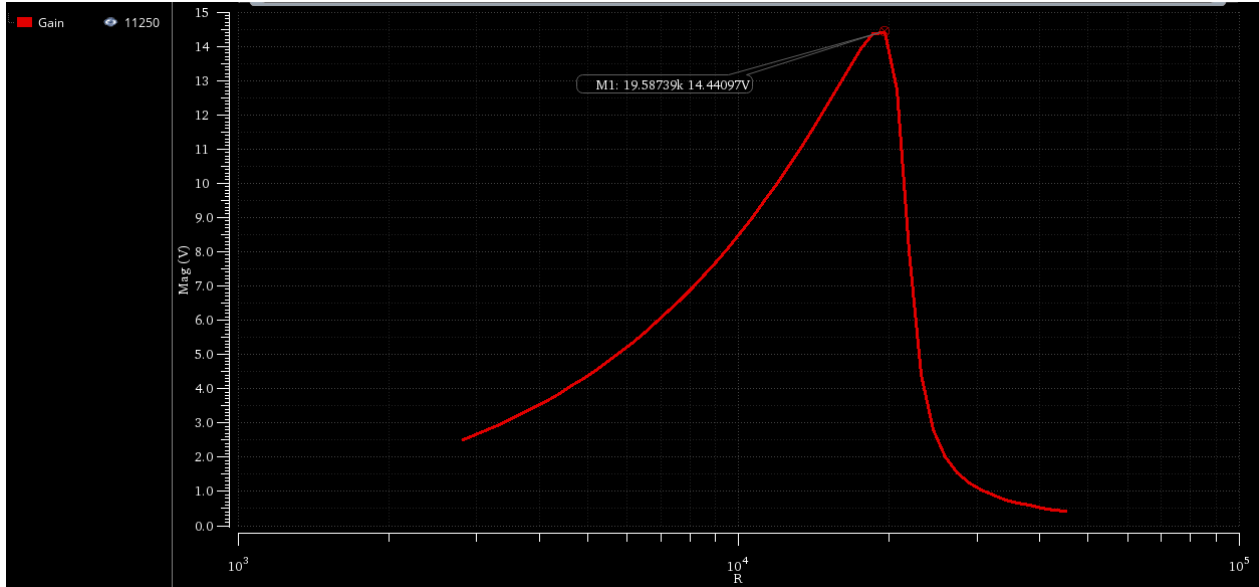


Figure 12. Gain vs. R_D sweep from AC analysis.

$$R_D^{(\text{sim})} = 19.587 \text{ k}\Omega \quad |A_v|_{\text{max}}^{(\text{sim})} = 14.44 \quad (13)$$

2.3.1 Hand Analysis: Gain as a Function of R_D

The small-signal gain including r_o is:

$$|A_v(R_D)| = g_m(R_D || r_o) = g_m \frac{R_D r_o}{R_D + r_o} \quad (14)$$

Note that **the bias current changes with R_D** when V_{GSQ} is held constant and the supply is fixed. In the simplified case where g_m and r_o are treated as constants:

- $R_D \ll r_o$: gain $\approx g_m R_D$ increases linearly with R_D .
- $R_D \gg r_o$: gain $\approx g_m r_o = A_{\text{int}}$, a constant.

In practice, increasing R_D eventually pushes V_{DS} below V_{ov} , causing the gain to *drop*. The maximum R_D just before saturation exit is:

$$R_D^{(\text{max})} = \frac{V_{DD} - V_{ov}}{I_D} = 20 \text{ k}\Omega \quad (15)$$

and the corresponding maximum gain:

$$|A_v|_{\text{max}} = g_m R_D^{(\text{max})} || r_o \approx \frac{2(V_{DD} - V_{ov})}{V_*} = 18 \quad (16)$$

Table 5. Simulation vs. hand analysis: maximum gain.

Parameter	Simulation	Hand Analysis	% Error
$R_D^{(\text{max})}$	19.587 kΩ	20 kΩ	2.065%
$ A_v _{\text{max}}$	14.44	18	19.7%

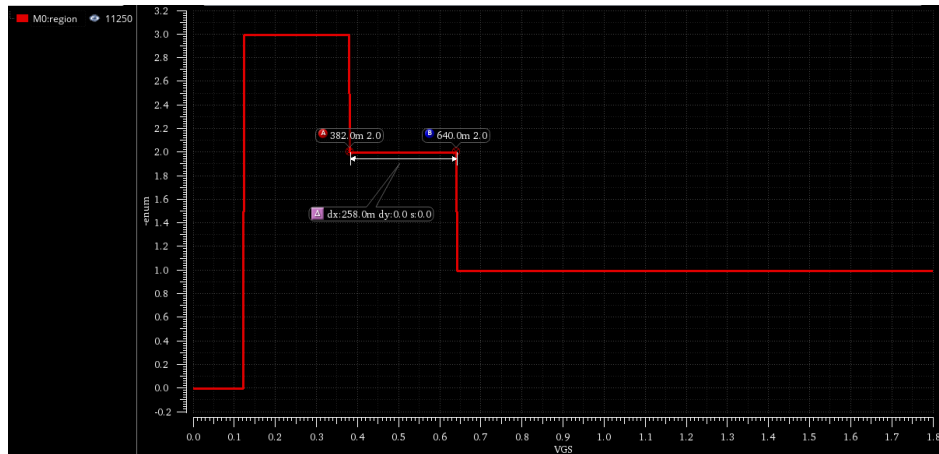


Figure 13. Region of NMOS vs. V_{GS} .

The available output swing equals **258 mV**.

Comments & Analysis

Why does gain first increase then decrease with R_D ?

- **Increasing phase** (R_D small): $R_D \ll r_o$, so increasing R_D directly increases $g_m R_D$. The transistor stays in saturation.
- **Decreasing phase** (R_D large): $V_{DS} = V_{DD} - I_D R_D$ falls below V_{ov} , pushing the transistor into triode. g_m drops sharply, g_{ds} rises, and the gain collapses.

Effect of supply scaling on gain:

Scaling down V_{DD} reduces the headroom available for the output swing. Since the maximum gain $\propto (V_{DD} - V_{ov})$, a lower supply directly reduces the achievable gain. This is a fundamental challenge in scaled CMOS technologies.

2.4 [Optional] Gain Linearization via Feedback

2.4.1 PMOS Active Load Schematic

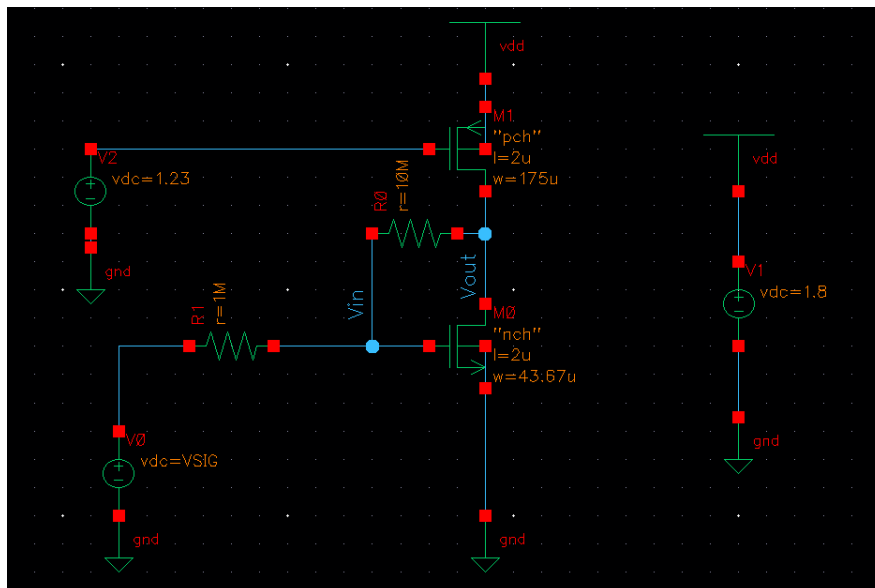
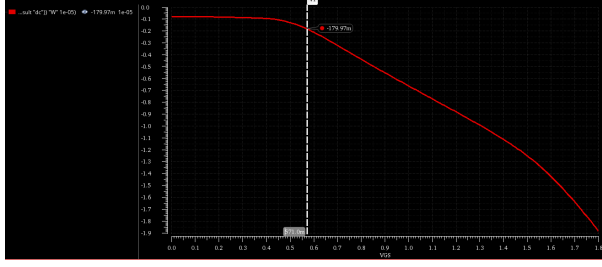
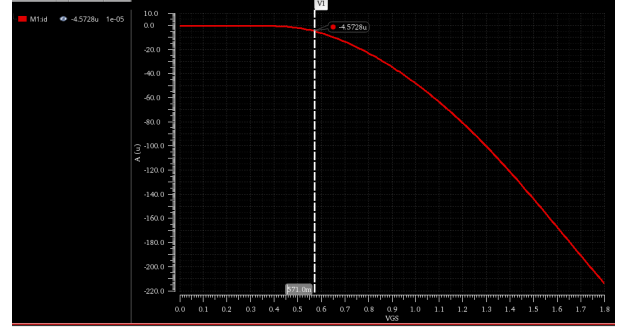


Figure 14. CS amplifier with PMOS current-source load and feedback resistors R_{in} and R_f .

2.4.2 PMOS Sizing Chart



(a) V^* vs. $|V_{GS}|$ for the reference PMOS.



(b) I_D vs. $|V_{GS}|$ for the reference PMOS.

Figure 15. Characterization curves for the reference PMOS ($W = 10\mu\text{m}$, $L = 2\mu\text{m}$).

From the PMOS characterization charts, we extract:

- $V_{GSX} = 571\text{ mV}$
- $I_{DX} = 4.5728\mu\text{A}$

2.4.3 PMOS Sizing

Using the same $V_Q^* = 0.18\text{ V}$ target and the required bias current $I_{DQ} = 80\mu\text{A}$:

$$W_P = 10\mu\text{m} \times \frac{I_{DQ}}{I_{DX,P}} \Big|_{V^*=0.18\text{ V}} = 10 \times \frac{80}{4.5728} = \boxed{175\mu\text{m}} \quad (17)$$

2.4.4 Feedback Resistors

The feedback resistor R_f sets the closed-loop gain:

$$|A_{v,\text{closed}}| \approx \frac{R_f}{R_{in}} \implies R_f = |A_v| \times R_{in} = 10 \times 1\text{ M}\Omega = 10\text{ M}\Omega \quad (18)$$

2.4.5 DC Sweep with Feedback

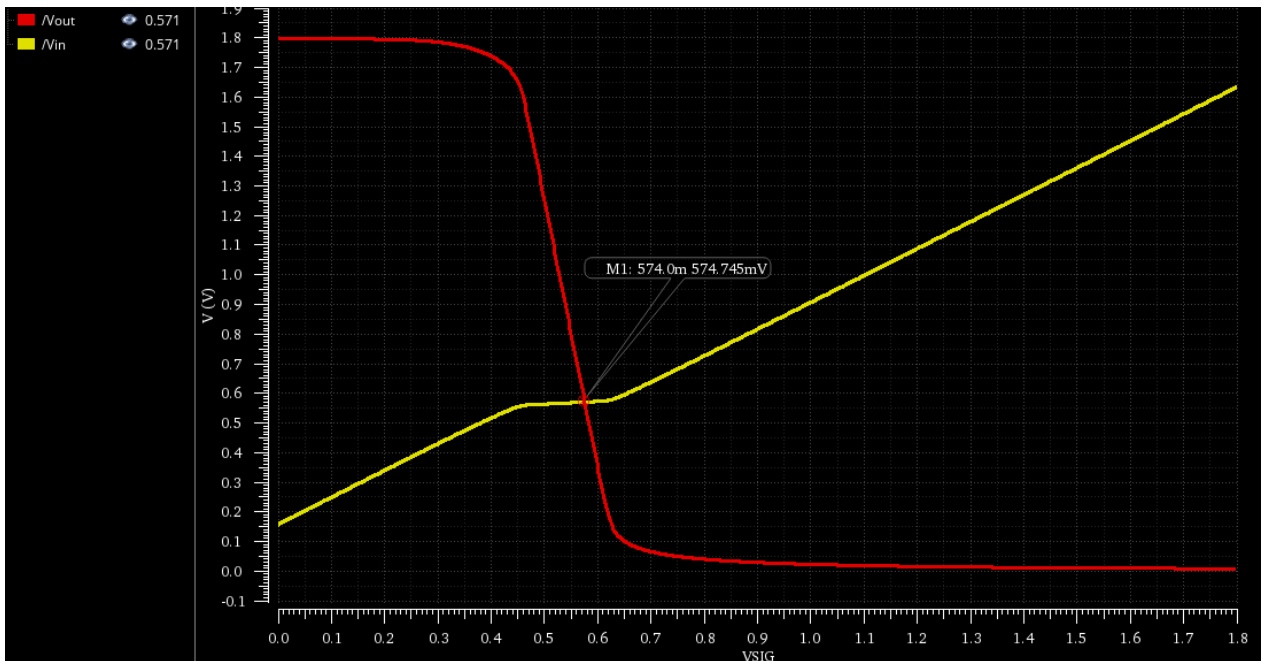


Figure 16. V_{IN} and V_{OUT} vs. V_{SIG} (DC sweep with PMOS load and feedback).

Comments & Analysis

At what voltage do the V_{IN} and V_{OUT} curves cross?

They cross at $V_{SIG} = 574 \text{ mV}$ (the DC bias gate voltage of M1), where $I_{DN} = I_{DP}$.

Is V_{OUT} vs. V_{SIG} more linear than before?

The relationship can be divided into three distinct regions:

- **Cut-off:** $V_{SIG} < V_{TH}$, so $V_{out} = V_{DD} \approx 1.8 \text{ V}$.
- **Saturation:** $V_{SIG} \geq V_{TH}$ and $V_{GD} \leq V_{TH}$. Feedback improves linearity; this is the useful amplification region with a high and approximately constant slope.
- **Triode:** $V_{SIG} > V_{TH}$ and $V_{GD} > V_{TH}$. Gain drops significantly.

The overall curve is *not* strictly linear across its full range, but the amplification region is far more linear than the open-loop case.

2.4.6 Derivative (Gain) with Feedback

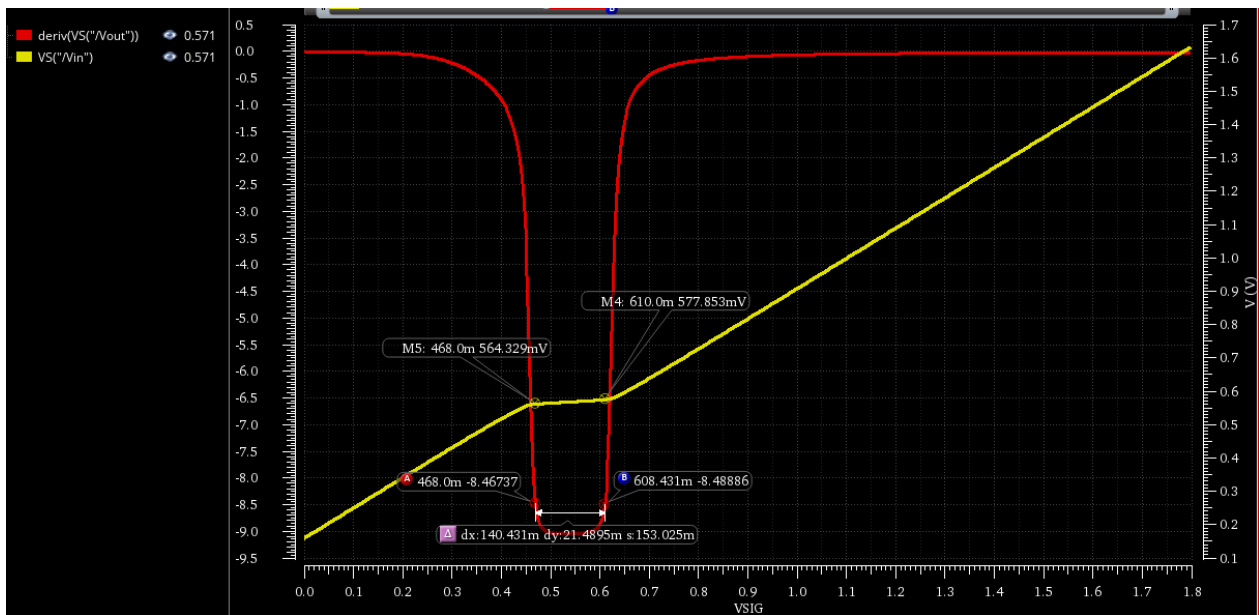


Figure 17. dV_{OUT}/dV_{SIG} (small-signal gain) vs. V_{SIG} with feedback.

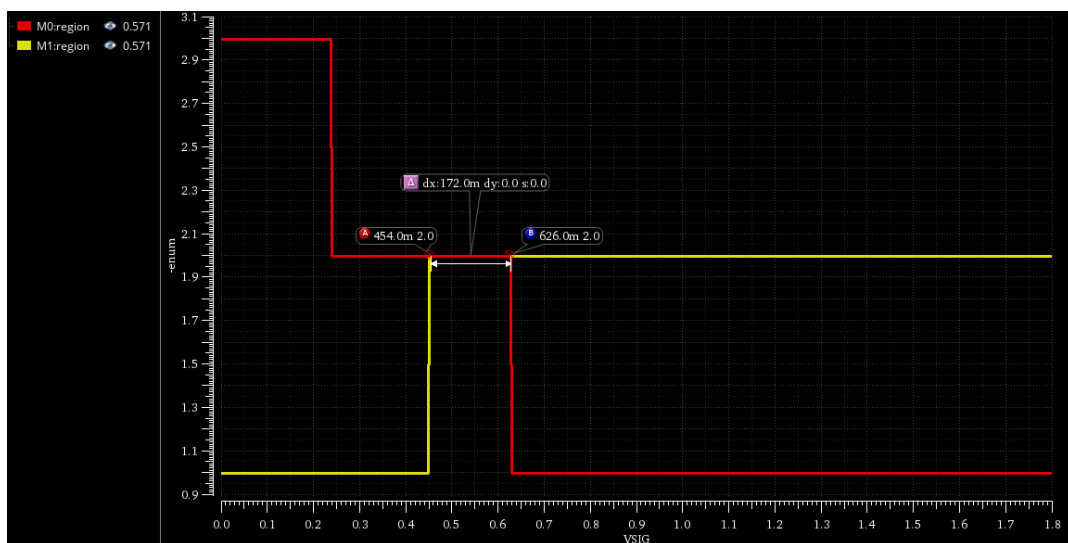


Figure 18. Regions of operation for NMOS and PMOS.



Is the gain more constant with feedback?

Yes. The derivative plot shows a much flatter gain characteristic compared to the open-loop case. Feedback desensitizes the closed-loop gain to g_m variations: as long as $A_{OL} \gg R_f/R_{in}$, the closed-loop gain $\approx R_f/R_{in}$ regardless of fluctuations.

Analytical linear input range:

The amplifier fails when M1 or M2 exits saturation ($V_{DS} < V^*$). Output range before clipping:

$$V_{DD} - 2V^* = 1.8 - 2(0.18) = 1.44 \text{ V} \quad (19)$$

$$\Delta V_{in} \approx \frac{V_{DD} - 2V^*}{|A_v|} = \frac{1.44}{10} = 144 \text{ mV} \quad (20)$$

For NMOS to be in saturation:

$$V_{GS} \geq 382.6 \text{ mV} \quad \text{and} \quad V_{out} \geq V_{ov} = V_Q^* = 180 \text{ mV}$$

$$V_{out} \geq 0.18 \text{ V}$$

For PMOS to be in saturation:

$$V_{SD} \geq V_{SG} - |V_{THP}|$$

$$V_{out} \leq V_G + |V_{THP}| \leq 1.2 + 0.393$$

$$V_{out} \leq 1.593 \text{ V}$$

So the range of V_{out} for linear gain is 1.413 V, giving:

$$\text{Range}_{V_{SIG}} = \frac{\text{Range}_{V_{OUT}}}{\text{Gain}} = \frac{1.413}{10} = 0.1413 \text{ V}$$

Using the formula:

$$\text{Range}_{V_{SIG}} = \frac{V_{DD} - 2V_Q^*}{|A_v|} = \frac{1.8 - 0.36}{10} = 0.144 \text{ V}$$

$$\text{Error}\% = \frac{0.144 - 0.134}{0.144} \times 100 = 6.944\%$$

The simulation shows the linear range for V_{SIG} spans 510 mV to 634 mV (134 mV), and for V_{IN} from 594.5 mV to 606.4 mV (11.9 mV), consistent with the analytical estimate.